

LendIQ

Credit Risk Analytics Pipeline

Project Writeup — End-to-End Documentation

LendIQ is an end-to-end credit risk analytics project built on 2.26 million real loan records from Lending Club (2007-2018). The project simulates the full workflow of a credit risk analyst at a fintech lender — from raw messy data through to a predictive default model — covering data engineering, SQL analytics, Power BI visualisation, and machine learning.

Dataset	Lending Club 2007-2018
Total Records	2,261,000 loans
Overall Default Rate	12.3%
Total Defaults	262,000 loans
Stack	Python, PostgreSQL, SQL, Power BI, LightGBM

Phase 1 — Data Engineering

The raw dataset arrived as a single 145-column CSV file. The first task was systematic assessment and cleaning to transform it into a structured relational database suitable for analytical queries.

Column Assessment & Reduction

Of the 145 original columns, a large proportion were either entirely null, redundant, or post-origination variables that would not be available at loan approval time (and therefore unusable for a predictive model). Columns were dropped according to the following criteria:

- More than 80% null values — insufficient data for meaningful analysis
- Post-origination variables — fields populated only after a loan is issued (e.g. recoveries, collection fees)
- Free-text fields — job title and description columns with no structured analytical value
- Duplicate information — columns encoding the same information in different formats

Schema Design & Normalisation

The cleaned dataset was normalised into four relational tables, each representing a distinct domain of information. This separation follows standard database design principles and makes queries more readable and maintainable.

borrowers	Borrower demographics — income, employment length, state, home ownership
loans	Loan terms — amount, grade, interest rate, purpose, issue date, loan status
credit_profile	Credit history — DTI, delinquencies, revolving balance, credit utilisation
payments	Payment behaviour — total payments received, recoveries, last payment amounts

All four tables are linked by a shared `loan_id` primary key, enabling joins across any combination of tables in a single query.

Data Type Fixes & Loading

Several columns required type conversion before loading into PostgreSQL. Interest rate was stored as a string with a percentage symbol and required parsing to float. Employment length was stored as text (e.g. '10+ years', '< 1 year') and was converted to a numeric representation. Issue date was converted from string to a proper date type to enable date truncation and extraction in SQL.

The cleaned and normalised data was loaded into PostgreSQL using Python and SQLAlchemy, and verified in DBeaver.

Phase 2 — SQL Analytics & Power BI Dashboard

With the data loaded into PostgreSQL, the analytical layer was built entirely in SQL — progressing from basic aggregations through to advanced window functions. Results were then visualised in a Power BI dashboard. The business question driving this phase was: which borrower and loan characteristics are most strongly associated with default?

Level 1 — Portfolio Overview

Basic aggregations established the shape of the portfolio. Loan volume is distributed across seven risk grades (A through G), with interest rates rising cleanly from around 7% for Grade A to over 25% for Grade G — consistent with risk-based pricing. The loan status breakdown revealed more complexity than expected:

- Fully Paid: 1,041,952 loans — clean repayment
- Current: 919,695 loans — still active, outcome unknown
- Charged Off: 261,655 loans — primary default label
- Late (31-120 days): 21,897 — pre-default, high risk
- Remaining statuses (Default, Grace Period, Late 16-30, credit policy exceptions): small volumes, treated as ambiguous

This status complexity was an early signal that the data would require careful handling before modelling.

Level 2 — Default Rate Analysis

The core metric throughout was the default rate, calculated using a CASE WHEN expression: loans with status 'Charged Off' or 'Default' were assigned 1, Fully Paid assigned 0, and the average taken per group.

Default Rate by Grade

The risk ladder is clear and monotonic. Grade A borrowers default at 3.18% while Grade G reaches 36.83% — more than ten times the rate. This confirms that Lending Club's internal grading system is a meaningful risk signal.

Default Rate by Loan Purpose

Small business loans carry the highest default rate at 18.08%. However, the more important finding comes from separating default rate from default count: debt_consolidation has a moderate rate of 12.60% but 161,058 actual defaults — by far the largest absolute loss driver in the portfolio. This distinction between probability of default (PD) and expected loss is a key credit risk concept.

DTI vs Default Rate

Borrowers were bucketed by DTI into four ranges. The relationship is clean and monotonic: Low DTI (0-10%) defaults at 8.69%, rising to 14.78% for Very High DTI (30%+). This confirms DTI as a useful predictive signal before any modelling begins.

Employment Length vs Default Rate

Employment length proved to be a weaker predictor overall. The middle range (1-9 years) shows no clear pattern, with all buckets clustered between 11.37% and 12.91%. The two notable exceptions are borrowers with unknown employment length (13.88% — the highest) and those with 10+ years employment (10.82% — the lowest). The missing employment flag turned out to be more informative than the raw value.

Level 3 — Window Functions

Advanced SQL was used to build time-series and comparative analytics using CTEs and window functions.

- Monthly default rate over time (LAG): default rates rose through the 2010-2015 period, peaking around 2015-2016 before dropping sharply in 2017-2018 — likely reflecting tighter underwriting standards in later vintages
- Rolling 3-month default rate by grade (AVG OVER): smoothed the monthly volatility and confirmed the grade separation is consistent over time, with Grade G tracking well above all other grades
- State ranking by default rate within grade (RANK OVER PARTITION BY): identified geographic concentration of risk within each grade
- Income decile analysis (NTILE): confirmed that lower income deciles default at higher rates, consistent with the DTI finding
- Vintage analysis: loans issued in 2015-2016 show the highest default rates, while 2017-2018 vintages perform significantly better

Power BI Dashboard

All SQL query outputs were loaded into Power BI and visualised across three dashboard pages. Key design decisions: aggregated query results were exported rather than the full 2.26M row dataset, keeping the file size manageable. The dashboard covers the grade risk ladder, purpose default rates, the DTI monotonic relationship, employment length breakdown, the LAG-based monthly trend, rolling 3-month rates by grade, state map, income decile chart, and vintage analysis line chart.

Phase 3 — Predictive Modelling (LightGBM)

The final phase exported the SQL-engineered features into Python and trained a LightGBM binary classifier to predict probability of default. The goal was not to build a production-ready model but to validate whether the analytical findings from Phase 2 translated into genuine predictive signal.

Target Variable Definition

The 9-way loan status was collapsed into a binary target. The ambiguous statuses (Current, In Grace Period, Late 16-30 days, credit policy exceptions) were dropped, leaving only resolved loans with a known outcome.

Label 1 (Default)	Charged Off + Default + Late (31-120 days) = 283,583 loans
Label 0 (Fully Paid)	Fully Paid = 1,041,952 loans
Final Dataset	1,325,535 loans — class ratio approximately 78/22

Feature Set

Nine features were selected based on the SQL analytical findings. Features known to be weak predictors were retained but expected to contribute little; one binary feature was engineered from a data quality observation.

int_rate	Interest rate — continuous, strong risk signal
loan_amnt	Loan amount — continuous
annual_inc	Annual income — continuous
dti	Debt-to-income ratio — validated monotonic signal in Phase 2
grade	Loan grade — categorical A-G
purpose	Loan purpose — categorical, 14 categories
issue_year	Year of loan issue — vintage signal
emp_length	Employment length in years — weak predictor
emp_length_missing	Binary flag: 1 if employment length was null — engineered feature

Preprocessing

Minimal preprocessing was required. The 77,391 null values in emp_length were filled with the column median. The 340 null values in dti were similarly median-filled. Grade and purpose were

cast to category dtype, which LightGBM handles natively without one-hot encoding — a key advantage of the algorithm for high-cardinality categoricals.

Model Training

The dataset was split into three sets: 70% training (928,404 rows), 15% validation (198,300 rows), and 15% test (198,831 rows). Stratified splitting was used to preserve the 78/22 class ratio across all three sets.

LightGBM was selected for three reasons: speed on large tabular datasets, native categorical support, and its prevalence in industry credit risk applications. `scale_pos_weight` was set to the ratio of negative to positive class (approximately 3.67) to account for class imbalance. Early stopping on validation AUC was used to prevent overfitting.

Results

Test AUC	0.7099
Classification Threshold	0.30 (adjusted from default 0.50 for imbalanced data)
Default Recall	45% — catches 45% of actual defaults
Default Precision	40% — when flagged as default, correct 40% of the time
Fully Paid Recall	81%

An AUC of 0.71 represents meaningful discriminative power above random (0.50) for a 9-feature model. The default classification threshold was lowered from 0.50 to 0.30, reflecting the asymmetric cost structure in credit risk — missing a default is more costly than incorrectly flagging a good loan.

Feature Importance

The feature importance chart confirmed and extended the Phase 2 findings:

- `int_rate` ranked first (762) — interest rate is the dominant predictor, which makes sense as it is Lending Club's own risk price signal, effectively a pre-computed risk score
- `loan_amnt` and `annual_inc` ranked second and third (576, 547) — together they approximate a DTI-style ratio
- `dti` ranked fifth (392) — consistent with the monotonic relationship identified in SQL
- `emp_length_missing` ranked sixth (227), outperforming raw `emp_length` (68) — validating the feature engineering decision
- `grade` and `purpose` ranked lower (94, 84) — `grade` is largely captured by `int_rate`; `purpose` adds marginal incremental signal
- `emp_length` ranked last (68) — consistent with the SQL finding that it is a weak predictor

Limitations & Next Steps

The model has a natural ceiling given the feature set. `int_rate` and `grade` are correlated (`grade` determines `rate`), limiting the independent signal available. A production credit risk model would incorporate bureau data (credit score, payment history, number of open accounts), behavioural features, and potentially macroeconomic variables — which would be expected to push AUC toward 0.80 or above.

The 45% default recall at a 0.30 threshold means the model misses over half of defaults. In a real lending context, the threshold would be calibrated to the lender's risk appetite and the relative cost of false negatives versus false positives.